

Midterm Exam 1 · Fall 2022

MATH 411 · October 7, 2022

Complete worked solutions

Retypeset past exam for practice. The original allowed one handwritten page of notes, lasted 50 minutes, and was worth 25 points.

1. (a) **(3 points)** Find every solution of $z^3 = -8i$. You may leave your answers in polar form $re^{i\theta}$.

Solution. Write $z = re^{i\theta}$. Since $-8i = 8e^{3\pi i/2}$, we have $r^3 = 8$ and $3\theta = 3\pi/2 + 2\pi k$. Thus $r = 2$ and the three distinct roots are

$$z_k = 2 \exp\left(i\left(\frac{\pi}{2} + \frac{2\pi k}{3}\right)\right), \quad k = 0, 1, 2.$$

- (b) **(4 points)** Let $r > 0$ be real and $w \in \mathbb{C}$. Show that

$$\left| \frac{w - r}{1 - r\bar{w}} \right| \leq 1 \quad \text{when } r < 1 \text{ and } |w| < 1.$$

Solution. The denominator is nonzero because $|r\bar{w}| < 1$. Compare the squares of the two moduli:

$$\begin{aligned} |w - r|^2 &= |w|^2 - r(w + \bar{w}) + r^2, \\ |1 - r\bar{w}|^2 &= 1 - r(w + \bar{w}) + r^2|w|^2. \end{aligned}$$

Thus $|w - r|^2 \leq |1 - r\bar{w}|^2$ is equivalent to

$$(1 - r^2)(1 - |w|^2) \geq 0.$$

Both factors are positive under the stated hypotheses. In fact the inequality is strict, so the quotient has modulus less than 1.

2. **(9 points)** Let $D = \{z \in \mathbb{C} : |z| < 1\}$ and let $f = u + iv$ be holomorphic on D . Suppose $|f|^2 = u^2 + v^2 = c$ throughout D , where $c > 0$. Prove that f is constant.

Hint. Differentiate $u^2 + v^2 = c$ with respect to x and y . You may use that a real function with continuous partial derivatives and both partial derivatives zero on D is constant.

Solution. Differentiating gives $uu_x + vv_x = 0$ and $uu_y + vv_y = 0$. The Cauchy–Riemann equations say $v_x = -u_y$ and $v_y = u_x$, so

$$\begin{pmatrix} u & -v \\ v & u \end{pmatrix} \begin{pmatrix} u_x \\ u_y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

The matrix has determinant $u^2 + v^2 = c > 0$. Hence $u_x = u_y = 0$; the Cauchy–Riemann equations also give $v_x = v_y = 0$. Both u and v are constant on the connected disc D , so f is constant.

3. **(9 points)** Let f be entire and analytic, and set

$$A_n = \{z \in \mathbb{C} : f^{(n)}(z) = 0\}, \quad n \geq 0.$$

- (a) Show that A_n is closed. Show also that A_n is discrete if $f^{(n)}$ is not identically zero. What is A_n if $f^{(n)}$ is identically zero?

Here *discrete* means that every point of the set has a neighborhood containing no other point of the set.

Solution. The derivative $f^{(n)}$ is continuous, so its zero set A_n is closed. If $f^{(n)}$ is not identically zero, the isolated zeros theorem makes each point of A_n isolated; thus A_n is discrete. If $f^{(n)} \equiv 0$, then $A_n = \mathbb{C}$, which is closed but not discrete.

- (b) Suppose that for every $a \in \mathbb{C}$, at least one coefficient in the Taylor expansion

$$f(z) = \sum_{n \geq 0} c_n(a)(z - a)^n$$

is zero. Show that f is a polynomial. You may use part (a) and the fact that $\mathbb{C}$ is uncountable.

Solution. Assume instead that f is not a polynomial. Then no $f^{(n)}$ vanishes identically, since $f^{(n)} \equiv 0$ would make f a polynomial. By part (a), each A_n is discrete and closed. Each discrete subset of $\mathbb{C}$ is countable: for each point choose a disc with rational center and rational radius that contains it but no other point of the subset. Distinct points require distinct such discs, and there are only countably many rational discs.

Now $c_n(a) = f^{(n)}(a)/n!$. The hypothesis therefore says that every a belongs to at least one A_n , so $\mathbb{C} = \bigcup_{n \geq 0} A_n$. The right side is countable, contrary to the uncountability of $\mathbb{C}$. Thus f is a polynomial.